Information-retention certificates for quota audits

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These notes state the exact scope of the supplementary study. General row-space query support and query auditing are established principles; the notes apply the existing quota/transfer-graph characterization and verify its consequences. All dimensions below concern a single scalar metric and universal exact linear representations on the declared atom space, unless a feasible source domain is explicitly stated. They are not minimum-bit or privacy theorems.

Model. For $G \geq 1$ and $K \geq 1$, let $Q = \{q \in \mathbb{Z}^G : \ell_g \leq q_g \leq u_g, \sum_g q_g = K\}$ be nonempty, with $0 \leq \ell_g \leq u_g \leq K$. A cumulative group/rank vector is $x \in \mathbb{R}^{GK}$ and $a_q(g,r) = \mathbf{1}[r \leq q_g]$. The rows of A are a_q, $q \in Q$. Let $m_g = \min_Q q_g$, $M_g = \max_Q q_g$. Mandatory positions satisfy $r \leq m_g$; variable positions V satisfy $m_g < r \leq M_g$; unreachable positions satisfy $r > M_g$. The transfer graph on V contains an edge $(g,r),(h,s)$ when legal neighboring quotas differ by adding (g,r) and removing (h,s). Its components are $C_1, \ldots, C_c$. Fix a legal base q_0.

Lemma 1 (Transfer span)**.** *Put $D = \{z : z = 0$ off V, $\sum_{v \in C_i} z_v = 0$ for every $i\}$. Then* $\mathrm{row}(A) = D + \mathrm{span}\{a_{q_0}\}$ *and* $\mathrm{rank}(A) = 1 + |V| - c$.

Proof. The legal quota graph is connected: if two quotas differ, transfer one unit from a coordinate above its destination value to one below its destination value. Bounds remain satisfied and the ℓ_1 distance decreases by two. Every transfer-row difference is a signed graph-edge difference. Conversely, each graph edge has a legal neighboring-quota witness, so its difference lies in $\mathrm{row}(A)$. Edge differences span exactly D, with dimension $|V| - c$. Connectivity of the legal quota graph now gives the claimed row-space equality. In every component the number $k_i = \sum_{v \in C_i} a_{q_0}(v)$ selected by a legal quota is invariant under transfers and is positive: otherwise a variable position in that component could never be selected. If mandatory positions exist, a_{q_0} has a nonzero mandatory coefficient and is outside D. Otherwise V is nonempty because $K > 0$, and some component has positive coefficient sum in a_{q_0}, also putting it outside D. The case $V = \varnothing$ has one nonzero quota row and rank one. □

Proposition 1 (Structural target test)**.** *A target $t^\top x$ has a universal linear decoder from all permitted answers iff there is one scalar α such that: (i) t vanishes at unreachable positions; (ii) every mandatory coefficient equals α; and (iii) $\sum_{v \in C_i} t_v = \alpha k_i$ for every component.*

Proof. The conditions say exactly that $t - \alpha a_{q_0} \in D$. Apply the lemma and the standard row-space criterion for exact linear query support. □

If the test succeeds, any implementation supporting all exact answers also supports this reconstruction by composing its query interface with the linear decoder; this implication does not require linear internal storage. If it fails, there is no universal linear decoder on the ambient atom space. Failure alone establishes no non-disclosure claim on a constrained feasible domain or at a particular observed answer vector.

Corollary 1 (Relative information can remain exposed)**.** *A zero-sum target t lies in* $\mathrm{row}(A)$ *iff it lies in D. In particular, the difference of two distinct reachable rank coordinates is recoverable iff both are variable positions in the same transfer component. The number of such recoverable pairwise contrasts is $\sum_i \binom{|C_i|}{2}$.*

Proof. In $t = z + \alpha a_{q_0}$, $z \in D$ has total coefficient sum zero, while a_{q_0} has total sum $K > 0$. Thus a zero-sum t forces $\alpha = 0$. Apply the component-sum description of D. □

Nonuniqueness of individual coordinates therefore does not imply that their differences or other sensitive linear combinations are hidden.

Proposition 2 (Additional retained linear information)**.** *For a linear state Ex supporting every row of A, its extra independent linear information is* $\operatorname{rank}([A;E]) - \operatorname{rank}(A) = \operatorname{rank}(E) - \operatorname{rank}(A)$*. The answer vector itself and a transfer-forest basis both have zero extra dimensions and the same kernel. Counter count can differ even when their independent information dimensions coincide.*

Proof. Support gives $\operatorname{row}(A) \subseteq \operatorname{row}(E)$. The rank difference is the dimension of the quotient of their row spaces. For the forest, the lemma gives equality of row spaces; equality of kernels follows. Thus each answer-equivalent state is a deterministic linear function of the other. No inference advantage over all-answer materialization follows from using the forest. □

M0 reconstruction. The checked M0 contract has no mandatory positions and one component containing 20 reachable positions. Its rank is therefore 20, the full reachable dimension. Tree-edge differences recover potentials $p_v = x_v - x_{v_0}$ from pairs of legal query answers. One base answer fixes the root as $x_{v_0} = (a_{q_0}x - \sum_{v:q_0 \text{ selects } v} p_v)/K$. At most $1 + 2(20-1) = 39$ distinct queries are needed before deduplication. The actual construction uses 28. Independently, 20 linearly independent legal rows form an invertible matrix on reachable positions and give a 20-query decoder. Twenty scalar linear answers per metric are necessary for universal linear recovery of 20 independent reachable coordinates. Both decoders use permitted answers alone; the separately captured source validates their outputs. Recovered quantities are cumulative group/rank metrics, not individual records.

Proposition 3 (Binary full-range kernel)**.** *For $G = 2$, $q = (i, K-i)$, $0 \le i \le K$, pair the coordinates $(0, r)$ and $(1, K+1-r)$. The answer rank is $K+1$, and the kernel consists exactly of vectors assigning a common offset z_r to both coordinates of pair r, with $\sum_{r=1}^{K} z_r = 0$. A basis is $d_r = p_r - p_K$, $r < K$, where $p_r = e_{0,r} + e_{1,K+1-r}$.*

Proof. Consecutive quota rows differ by $e_{0,r} - e_{1,K+1-r}$. Their orthogonality conditions force equal offsets inside each pair. The base quota selects one coordinate of every pair, so its condition is $\sum_r z_r = 0$. These $K-1$ independent degrees of freedom give rank $2K - (K-1) = K+1$. □

Fixed-catalogue feasible-domain certificate. The source domain holds the 62,423-movie catalogue fixed and uses MovieLens half-star ratings, its parsed release-year groups and metrics, score-descending and movie-ID-ascending ranks, and one sealed cumulative release horizon. Around the observed 25M-rating history, the deterministic search finds, for each $r = 1, \ldots, 9$, four movie substitutions whose cumulative year-metric change is d_r. It uses 36 distinct users across all directions. Every replacement is a catalogue member absent from that user's history; its release year changes by the required ± 1, stays in the same binary group, and is no later than the rating's timestamp year. The rating, timestamp, number of ratings and ranking position are preserved. Each original and modified user history is independently sorted and directly queried to verify its exact contribution. No catalogue attribute is changed, and modified preferences are explicitly counterfactual rather than claimed observed behavior.

Because no edited user is shared by two directions, any subset combines independently validated user histories without interactions. The $2^9 = 512$ subsets are valid source histories with the same 22 policy/metric answers. Their year-atom differences have affine dimension nine, exhausting the year-metric kernel. Every one of the 20 individual year-rank coordinates varies across these histories at the same observed answer vector. A second SQL implementation re-parses and re-sorts all 25,000,095 records after all 36 edits and verifies the resulting aggregates and all permitted answers.

Corollary 2 (Exact source-domain identifiability at the observed answers)**.** *At this observed joint answer vector, a linear statistic of the 20 year atoms is uniquely determined over the declared fixed-catalogue source domain iff its coefficient vector lies in* $\operatorname{row}(A)$*.*

Proof. Membership gives a decoder on every feasible history. For nonmembership, the coefficient vector is not orthogonal to $\ker(A)$. Since the nine certified feasible differences span that kernel, it has nonzero inner product with some d_r. The observed history and that valid switched history have the same answers for both metrics and different target values. Thus even a nonlinear decoder cannot uniquely determine this linear target at these answers. □

This is constructive nonuniqueness for a specified domain and release, not a probabilistic privacy guarantee, a frequency estimate, or a guarantee under arbitrary auxiliary knowledge. Intermediate user-prefix answer releases can distinguish these histories; the experiment explicitly checks this limitation. Rating-metric values are unchanged, so the source certificate does not claim 18 feasible independent directions across both metrics.

Implicit interval compiler. Effective minima and maxima are $m_g = \max(\ell_g, K - \sum_{h \neq g} u_h)$ and $M_g = \min(u_g, K - \sum_{h \neq g} \ell_h)$. For a group pair $g < h$ and receiver rank r, feasible donor ranks form one interval obtained by intersecting $[m_h + 1, M_h]$ with $[K - r + 1 - \sum_{j \neq g,h} u_j,\ K - r + 1 - \sum_{j \neq g,h} \ell_j]$. Every integer residual in this interval is attainable by filling the remaining bounded coordinates greedily.

The implementation adds the receiver's edge to the first donor in an interval. It also adds an edge to each donor immediately after a donor-adjacency cut that has not yet been processed for this group pair. A successor structure skips processed cuts. A processed cut certifies that its consecutive donors are already connected by a previously examined star. Inductively each examined star's entire donor interval becomes connected to its receiver, while every edge actually added is a feasible original transfer edge. The connected components are therefore identical to a dense edge scan. Union-find discards cycles, producing a valid transfer forest.

There are $O(G^2K)$ interval/cut operations, $O(GK)$ stored vertices/forest edges, and $O(K)$ temporary successor entries per group pair. These are operation counts, with union-find overhead and integer bit costs kept explicit. Exact bounded-composition counting uses $O(GK)$ arithmetic operations. Exhaustive small-box checks compare components' resulting row spaces, forest witnesses, ranks and target decisions with independent rational elimination. Timing comparators include a dense structural scan, explicit matrices where within declared caps, and direct analytic formulas on their applicable families; the generic compiler is not claimed faster than those specialized formulas.